

TACTILE: Task-Aware Cascaded Tensor Inference for Local Edge

DVCon India 2026 Design Contest Stage 1 Report

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List of Symbols

TACTILE	— Task-Aware Cascaded Tensor Inference for Local Edge
OD-IP	— Object Detection Intellectual Property block (YOLOv5n engine)
SAM-IP	— Spatial task Attention Map Intellectual Property block
NMS-IP	— Non-Maximum Suppression Intellectual Property block
GOPS	— Giga Operations Per Second
MACs	— Multiply-Accumulate operations
INT8	— 8-bit integer quantization
FPS	— Frames Per Second
BRAM	— Block RAM (on-chip memory resource in Xilinx FPGAs)
DSP	— Digital Signal Processor (DSP48E1 slice on Kintex-7)
AXI	— Advanced eXtensible Interface (ARM AMBA bus protocol)
DDR3	— Double Data Rate 3 SDRAM
CDC	— Clock Domain Crossing
LUT	— Look-Up Table (basic FPGA logic element)
RoI	— Region of Interest
GAP	— Global Average Pooling
PTQ	— Post-Training Quantization
MIG	— Memory Interface Generator (Xilinx DDR controller IP)
MMCM	— Mixed-Mode Clock Manager
WFI	— Wait For Interrupt (RISC-V instruction)
mAP	— mean Average Precision
COCO	— Common Objects in COntext (dataset)

Abstract

This report presents TACTILE (Task-Aware Cascaded Tensor Inference for Local Edge), a hardware-software co-designed edge AI inference system for task-conditioned object detection, submitted to the DVCon India 2026 Design Contest. Standard object detectors are task-agnostic: they report all detected objects with equal priority, regardless of the user's current goal. TACTILE addresses this limitation by integrating task context at the earliest possible stage of the detection pipeline, exploiting the fact that the task is known before inference begins.

The system is deployed on the Digilent Genesys-2 FPGA board (Xilinx Kintex-7 XC7K325T) using the VEGA AS1061 RV64GC RISC-V soft processor as system controller. TACTILE's programmable logic (PL) domain comprises three custom IP blocks: OD-IP, a YOLOv5n INT8 inference engine built around an 8×8 systolic array delivering 19.2 GOPS; SAM-IP, a novel Spatial task Attention Map network that generates a 40×40 binary spatial mask prior to backbone inference at zero latency cost; and NMS-IP, a task-score-fused Non-Maximum Suppression engine that replaces confidence-only ranking with a three-way fused score combining detector confidence, task relevance, and class-task prior probability.

The system accepts a 160×160 RGB image and a 4-bit task ID (drawn from 14 COCO-Tasks) and outputs the bounding box and class of the most task-relevant object. TACTILE targets a minimum of 50 FPS at 160×160 resolution within a 50,000 LUT slice budget and under 5 W total system power. Theoretical analysis yields approximately 77 FPS (13.0 ms/frame), with a conservative hardware estimate of approximately 56 FPS (18 ms/frame) accounting for DDR3 arbitration and control overhead. The design consumes an estimated 33,880 slices (67.8%), 194 DSP48E1 (23.1%), and 421 BRAM36 (94.6%), all within contest resource limits.

I. Introduction

Object detection is a fundamental computer vision task that has achieved remarkable accuracy through deep neural networks such as YOLO, SSD, and Faster R-CNN. However, all of these systems are inherently task-agnostic: given a kitchen scene, they report every detectable object (knife, fork, wine glass, beer glass, cup, plate) with equal priority. This is appropriate for general-purpose perception, but inadequate for robotic or assistive applications where the system must identify the single most appropriate object for a given goal.

The COCO-Tasks benchmark (Sawatzky et al., CVPR 2019) formalizes this problem by pairing 14 household tasks (e.g., "serve wine", "smear butter", "dig a hole") with image-level annotations indicating which object a human would choose to accomplish that task. The task is specified as a short text description, which the system must use to filter and prioritize its detections.

Deploying a task-driven detection system on an FPGA-based edge device poses two challenges beyond standard object detection. First, task context must be integrated into the inference pipeline without adding unacceptable latency. Naively running full detection followed by task-based ranking wastes compute on task-irrelevant regions. Second, the FPGA resource budget is tight: the Kintex-7 XC7K325T provides 50,750 logic slices, 840 DSP48E1 units, and 445 BRAM36 tiles, which must accommodate both the VEGA soft processor and all custom accelerator logic.

TACTILE addresses both challenges through a co-designed pipeline. The key architectural innovation is task-aware spatial pre-filtering: a lightweight Spatial task Attention Map (SAM-IP) runs in parallel with the image DMA load, consuming zero inference cycles, and identifies image regions likely to contain task-relevant objects before backbone computation begins. This spatial guidance then flows through the entire pipeline, influencing confidence scoring at the RoI extraction stage and providing the spatial prior for the fused NMS stage. The result is a system that is not merely a detector with post-hoc task filtering, but a genuinely task-conditioned inference engine.

II. Background Research

Task-Driven Object Detection

The foundational reference for this work is Sawatzky et al. (2019), "What Object Should I Use? — Task Driven Object Detection" (arXiv:1904.03000). This paper introduces the COCO-Tasks dataset, which associates 14 manipulation tasks from the COCO dataset with preferred object annotations. The paper proposes a TASKNet architecture that combines standard detection with a task-conditioned scoring module, demonstrating that task context dramatically changes which object should be surfaced even when high-confidence detections exist for multiple object classes. TACTILE adopts the COCO-Tasks evaluation protocol and uses the same 14 task IDs.

Lightweight Edge Detection Backbones

Several lightweight detection networks were evaluated for FPGA deployment. YOLOv5n (Jocher et al.) was selected as the backbone for its combination of high representational efficiency per MAC (140M MACs at 160×160 input), standard 3×3 and 1×1 convolutions that map naturally to the systolic array without custom routing hardware, and an active open-source ecosystem with well-characterized INT8 post-training quantization. Alternatives including NanoDet-Plus (ShuffleNetV2 backbone), MobileNetV3 (depthwise convolutions), and a custom TASKNet-FPGA backbone were analyzed and rejected on quantifiable hardware efficiency grounds (see Section IV.i).

INT8 Quantization for FPGA

Post-training quantization (PTQ) of CNN weights and activations to 8-bit signed integers is standard practice for FPGA deployment. On the Kintex-7 DSP48E1, INT8 multiplication can be packed two-per-DSP by exploiting the 18-bit input width: one INT8 value in bits [7:0] and a second in bits [17:9]. This doubles effective throughput from 840 to 1,680 effective INT8 MACs per device at no additional resource cost. INT4 quantization was evaluated but rejected because Kintex-7 DSP48E1 has no native INT4 support; bit-serial LUT-based implementations consume more slices than INT8 for equivalent throughput.

Systolic Arrays for CNN Acceleration

Weight-stationary systolic arrays are the dominant architecture for CNN acceleration in area-constrained environments (Kung, 1982; Chen et al., 2016). In a weight-stationary dataflow, filter weights are preloaded into PE registers and held fixed while input activations flow through the array, maximizing weight reuse. This is particularly efficient for YOLOv5n, whose many spatial positions sharing the same filter weights produce high reuse factors. TACTILE's 8×8 INT8 systolic array with DSP48E1 packing achieves 19.2 GOPS sustained throughput.

VEGA AS1061 RISC-V Processor

VEGA AS1061 is an RV64GC soft processor IP developed by CDAC Trivandrum, implementing the full RISC-V 64-bit ISA with standard extensions (integer multiply/divide M, atomic A, compressed C, single/double FPU F/D). The M-extension provides hardware multiplication in one clock cycle, enabling the dot-product task scoring loop to run at

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approximately 50M MACs/second, sufficient to score 75 proposals in 0.19–0.31 ms. VEGA's WFI (Wait For Interrupt) instruction allows near-zero power consumption between inference frames, freeing DDR3 bandwidth for weight streaming.

III. Goal and Objectives

Goal

Design and deploy TACTILE, a task-conditioned object detection system that accepts a natural language task description (encoded as a 4-bit integer from 14 COCO-Tasks) and a camera image as paired inputs, and outputs the bounding box and class of the most task-appropriate object in the scene, running at a minimum of 50 FPS on the Digilent Genesys-2 FPGA board within the contest resource constraints.

Objectives

- Implement a YOLOv5n INT8 inference engine (OD-IP) using an 8×8 weight-stationary systolic array on the Kintex-7 XC7K325T, achieving 19.2 GOPS sustained throughput.
- Design a novel Spatial task Attention Map IP (SAM-IP) that generates a 40×40 binary spatial mask from the task ID and image thumbnail in parallel with image DMA, at zero marginal inference latency.
- Implement a task-score-fused NMS IP (NMS-IP) that ranks proposals by a three-way fused score (detection confidence × task relevance × class-task prior) rather than confidence alone.
- Integrate VEGA AS1061 as system controller, implementing the task scoring dot-product loop and firmware state machine in software for efficient resource partitioning.
- Achieve SAM recall $\geq 97\%$ per task on COCO-Tasks validation set through per-task threshold calibration.
- Meet all contest resource constraints: $\leq 50,000$ LUT slices, ≤ 430 BRAM36 (post-ILA removal), $F_{max} \geq 150$ MHz on `clk_pl`, ≤ 5 W total power.
- Verify correct operation on hardware using the `OD_STALL_CTR` register to confirm DDR3 weight streaming overlap and the fused NMS test case (wine glass preferred over beer glass for task "serve wine").

IV. Design Process

i. Problem Statement

Standard object detectors are task-agnostic. Given a kitchen scene, a conventional detector such as YOLOv5n reports all detected objects (wine glass, beer glass, knife, fork, cup, plate) with confidence scores reflecting only how certain the network is that an object of each class exists, with no regard for which object is useful for the current task. When the goal is "serve wine", the wine glass must be surfaced even if a beer glass has higher detector confidence.

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The central problem is therefore: how to integrate task context into the object detection pipeline so that the system consistently surfaces the most task-relevant object, while remaining deployable on a resource-constrained FPGA edge device with a RISC-V soft processor, and meeting a minimum throughput requirement of 50 FPS.

This problem has three sub-problems. First, the task context must influence detection at the earliest possible stage to avoid wasting compute on task-irrelevant regions. Second, task integration must not introduce unrecoverable errors: if the task-aware component makes a mistake, the pipeline must have a mechanism to self-correct. Third, the entire system — detection backbone, task integration, and NMS — must fit within 50,000 LUT slices and 445 BRAM36, with all critical paths meeting 150 MHz timing.

ii. Functional Specification

TACTILE accepts the following inputs per inference frame:

- A 160×160 RGB image stored in DDR3 SDRAM at a VEGA-specified address.
- A 4-bit task ID (0–13) corresponding to one of 14 COCO-Tasks, written by VEGA to the SAM_TASK_ID register before inference starts.

TACTILE produces the following outputs:

- Up to 5 surviving bounding box proposals, each encoded as: [63:48] fused score, [47:32] COCO class ID, [31:0] bounding box coordinates.
- Results are available in the NMS_RESULT_0 through NMS_RESULT_4 registers and transmitted over UART by VEGA firmware.

Key performance and accuracy specifications (Targeted):

Parameter	Specification
Input resolution	160×160 pixels, RGB uint8
Task encoding	4-bit integer, 14 COCO-Tasks
Target FPS	≥ 50 FPS (measured on hardware at 160×160)
Detection backbone	YOLOv5n, INT8 post-training quantization
SAM recall	≥ 97% per task on COCO-Tasks validation set
End-to-end mAP degradation	≤ 3% vs YOLOv5n baseline at 160×160
FPGA device	Xilinx Kintex-7 XC7K325T-2FFG900C (Genesys-2)
Soft processor	VEGA AS1061 RV64GC @ 100 MHz
PL clock	150 MHz (clk_pl)
Total system power	< 5 W
Slice budget	≤ 50,000 LUT slices
BRAM budget	≤ 430 BRAM36 (post-ILA removal)

iii. Proposed Design

System Architecture Overview

TACTILE's hardware-software partitioning follows the principle: accelerate operations that are compute-bound and data-parallel in programmable logic; execute operations that are control-bound or involve branching decisions on VEGA. Three custom IP blocks implement the acceleration pipeline, all clocked at 150 MHz in the PL domain:

- **OD-IP:** The YOLOv5n INT8 inference engine, comprising a preprocess unit, 8×8 weight-stationary systolic array, ping-pong activation buffers, weight streaming DMA, RoI align, global average pooling, and $256 \rightarrow 128$ -D feature projection. OD-IP is the primary compute block.
- **SAM-IP (Novel Contribution #1):** The Spatial task Attention Map network, a three-layer lightweight CNN (Conv $3 \times 3 \rightarrow$ Depthwise Conv $3 \times 3 \rightarrow$ Pointwise Conv $1 \times 1 \rightarrow$ bilinear upsample) with only 199 INT8 weight bytes stored as LUTRAM constants. SAM-IP runs in parallel with the OD-IP image DMA and completes before backbone inference begins, contributing zero cycles to the critical path.
- **NMS-IP (Novel Contribution #2):** The task-score-fused Non-Maximum Suppression engine that scores each proposal as $(\text{detection_confidence} \times \text{task_score} \times \text{class_task_prior})$ before IoU-based suppression. This three-way fusion ensures task-relevant objects consistently survive NMS even when task-irrelevant objects have higher raw detector confidence.

The VEGA AS1061 processor serves as system controller and implements the task embedding dot-product scoring loop (9,600 MACs over 75 proposals, approximately 0.31 ms worst case) and all result formatting and UART output logic. VEGA spends most of each inference frame in WFI, which both saves power and releases the DDR3 bandwidth exclusively for weight streaming.

Hardware/Software Partitioning

Function	Location	Justification
YOLOv5n convolution (all layers)	PL — OD-IP systolic array	140M+ MACs per frame, massively data-parallel
Spatial task pre-filtering (SAM-IP)	PL — SAM-IP	Runs parallel to DMA, zero latency cost
RoI align + feature projection	PL — OD-IP	Per-proposal spatial math, better in hardware
Task-fused NMS	PL — NMS-IP	Hardware novelty claim, 430 slices, simple RTL
Task scoring (dot product)	VEGA software	9,600 MACs, ~0.31 ms, not worth hardware overhead

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Function	Location	Justification
Result formatting and UART output	VEGA software	Control-bound, sequential
DDR weight loading at boot	VEGA firmware (DMA)	One-time setup operation

Figure 1: TACTILE System Top-Level Architecture

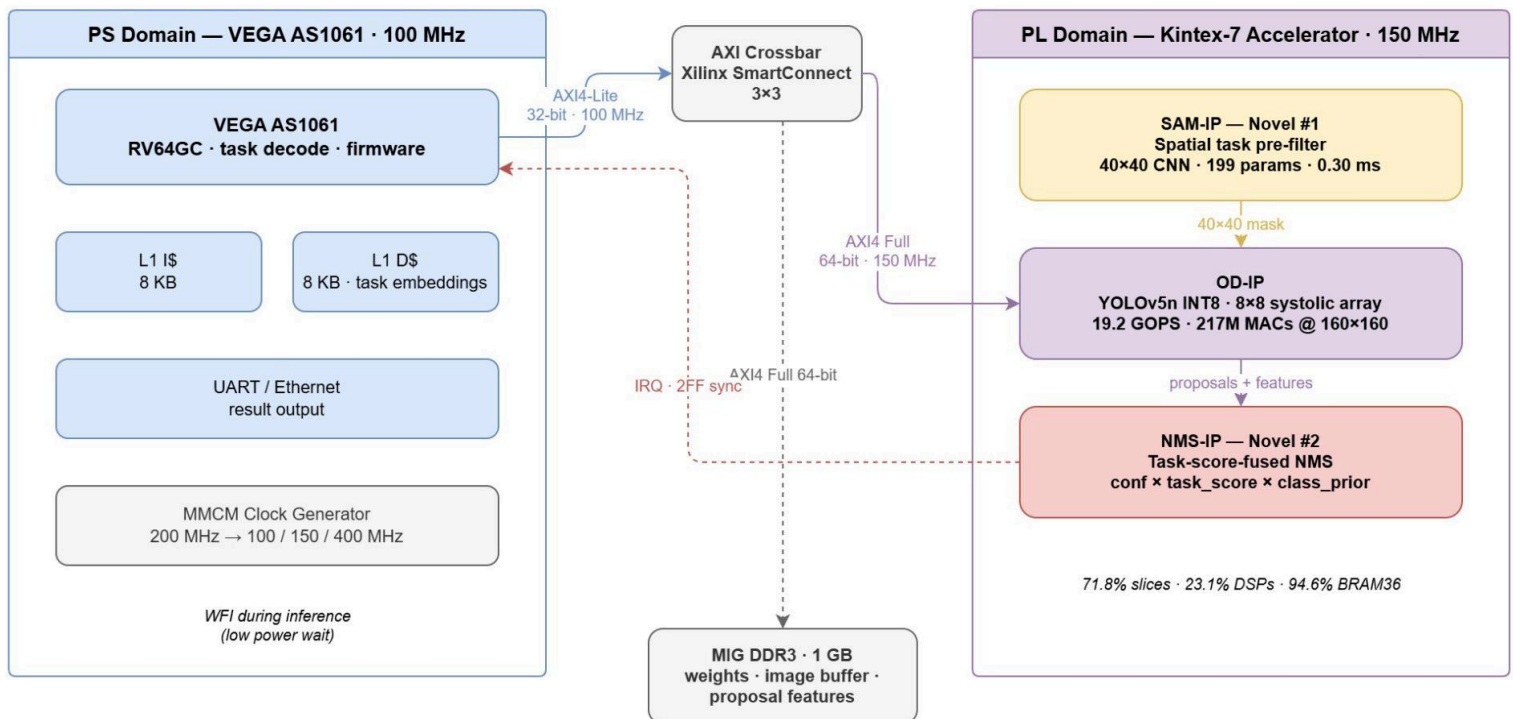
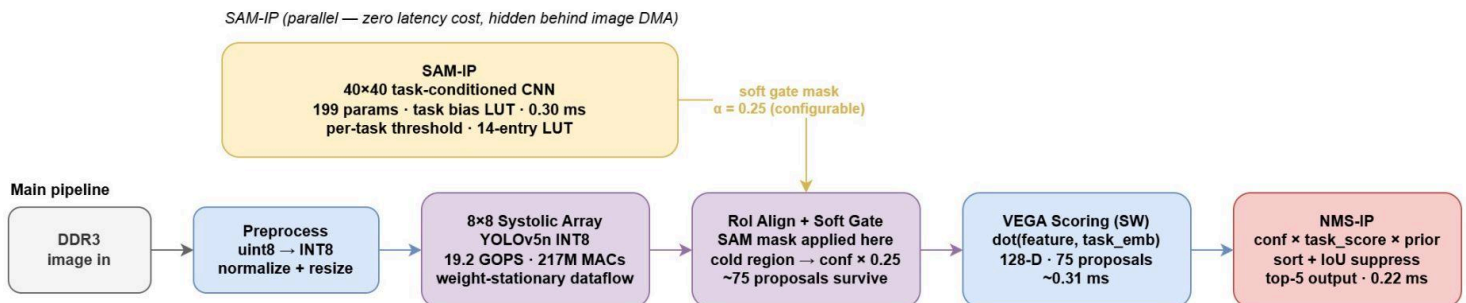


Figure 2: OD-IP Pipeline and Dataflow



Total: ~11 ms compute (160x160) + 0.31 ms VEGA scoring + 0.22 ms NMS-IP = ~18 ms conservative → ~56 FPS

OD-IP: YOLOv5n Inference Engine

The 8×8 systolic array uses weight-stationary dataflow: weight tiles are preloaded from BRAM into PE registers, then input activations stream left-to-right across rows while partial sums accumulate top-to-bottom through columns. Each DSP48E1 packs two INT8 MACs using the 18-bit input port (weight in bits [7:0], second activation in bits [17:9]), giving 128 effective INT8 MACs per cycle and 19.2 GOPS at 150 MHz. A requantization unit (4 DSP48E1) rescales 32-bit accumulators back to INT8 per output channel after each tile pass.

Weight memory is organized into four BRAM banks. Stem, Stage 1, Stage 2, PANet Neck, Detection Head, and SPP block weights (totalling 248 BRAM36, approximately 1.12 MB) are stored permanently in on-chip BRAM. Stage 3 and Stage 4 weights — the two largest layers — are stored in DDR3 and streamed into a single 86-BRAM36 reusable streaming buffer during the preceding computation stage. This weight streaming is fully overlapped: Stage 3 weights (0.30 ms) load during Stage 1+2 compute (3.0 ms, 10× margin), and Stage 4 weights (0.69 ms) load during Stage 3 compute (2.3 ms, 3.3× margin). Activation ping-pong buffers (25+25 BRAM36) eliminate stalls between consecutive layers.

After backbone inference, each surviving proposal is processed through SAM gate integration, bilinear RoI align (7×7 output), global average pooling (256-D), and a 1×1 convolution projection (256→128-D) using 8 BRAM18. This yields a 128-byte INT8 feature vector per proposal that VEGA uses for dot-product task scoring.

SAM-IP: Novel Contribution #1

SAM-IP generates a 40×40 binary spatial mask identifying image regions likely to contain task-relevant objects. It receives an 8× spatially downsampled 40×40×3 thumbnail sampled from the image DMA stream and a 3-byte task bias vector from a 14-entry LUTRAM table indexed by task_id. The three-layer CNN (112 + 40 + 5 = 157 weight bytes plus 42 task bias bytes = 199 bytes total) is implemented entirely in LUTRAM and DSP48E1 slices with no BRAM36 cost.

The output 10×10 logit map is bilinearly upsampled to 40×40 and thresholded by a per-task INT8 threshold from a 14-entry calibrated lookup table. Tasks with large, prominent target objects (e.g., "serve wine") use more aggressive thresholds; tasks with small or diverse targets (e.g., "dig a hole") use more permissive thresholds. Default pre-calibration values are 0x2D (approximately 0.35 normalized).

A critical design decision is the use of soft gating rather than hard-drop gating. Hard-drop (discarding proposals whose center falls in a cold mask region) creates unrecoverable false negatives that cannot be corrected by downstream scoring. Soft gating multiplies the effective confidence of cold-region proposals by SAM_ALPHA (default 0.25, Q0.8 format, configurable via AXI-Lite register) rather than dropping them. If VEGA's dot-product scoring assigns high task relevance to a soft-gated proposal, the high task score can overcome the confidence penalty; a hard-dropped proposal has no such recovery path.

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NMS-IP: Novel Contribution #2

Standard NMS ranks proposals by detector confidence alone, which is task-agnostic. NMS-IP replaces this single-criterion ranking with a three-way fused score:

$$\text{fused_score} = \text{detection_confidence} \times \text{task_score} \times \text{class_task_prior}$$

where task_score is the normalized dot-product computed by VEGA (INT8 Q0.8), and class_task_prior is a precomputed probability drawn from an 80×14 INT8 lookup table (1,120 bytes, synthesized as LUTRAM) indexed by COCO class ID and task ID. The mathematical basis is a factored probability model: under the independence assumption that the detector is task-agnostic and the scorer is confidence-agnostic, the probability of a correct task-driven detection factors as $P(\text{correct}|\text{task}) \approx P(\text{is_object}) \times P(\text{task_relevant}|\text{is_object}) \times P(\text{correct_class}|\text{task})$.

The worked example illustrates the necessity of fused scoring: for task "serve wine", a beer glass with detector confidence 0.81 beats a wine glass with confidence 0.72 under standard NMS. Under NMS-IP's fused scoring, the wine glass achieves $0.72 \times 0.94 \times 0.94 = 0.636$ versus the beer glass's $0.81 \times 0.18 \times 0.12 = 0.017$, correctly surfacing the wine glass. The pipeline includes a five-stage hardware implementation: proposal buffering, fused score computation (150 DSP cycles), bubble sort (2,812 comparisons), division-free IoU suppression ($2 \times \text{intersection} > \text{union}$ equivalence), and top-5 output register fill. Total NMS-IP latency is approximately 0.22 ms.

Clock and Reset Architecture

Three clock domains are generated by a single MMCM from the 200 MHz board oscillator: clk_vega at 100 MHz for the VEGA processor and AXI-Lite bus; clk_pl at 150 MHz for all three IP blocks and the AXI-Stream fabric; and clk_ddr at 400 MHz for the MIG DDR3 PHY, which is fully isolated. All cross-domain signals use documented CDC methods: AXI-Lite SmartConnect (with built-in synchronization) for VEGA-to-PL configuration, 2-flip-flop synchronizers for PL-to-VEGA interrupts (IRQ_SCORING and IRQ_DONE), and MIG internal async FIFOs for PL-to-DDR transfers. The PL domain is held in reset until VEGA firmware completes DDR3 calibration verification and weight loading.

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iv. Analysis of Final Design

Resource Budget (Tentative)

Module	Slices	DSP48E 1	BRAM3 6	Notes
VEGA AS1061	10,000	8	10	Soft processor + caches + ROM
MIG DDR3 Controller	3,000	0	4	Xilinx IP, fixed
AXI SmartConnect + DMA	1,500	0	8	Xilinx IP, fixed
OD-IP systolic array	13,000	128	248	8×8 MAC + permanent weights
OD-IP control + preproc	2,000	19	28	FSM + normalize + activation buffers
OD-IP RoI align + proj	2,000	4	4	Bilinear + 1×1 conv
OD-IP stream buffer	0	0	86	Stage 3+4 streaming, reusable
SAM-IP	1,450	13	1	Tiny CNN + mask register
NMS-IP	430	6	1	Sort + IoU + output registers
Clock/reset/GPIO	500	0	0	MMCM, board IO
TOTAL (no ILA)	33,880	194	421	67.8% / 23.1% / 94.6% of budget
BUDGET	50,000	840	445	Contest resource limits

Latency and Performance Model

Pipeline Stage	Compute (ms)	DDR Load (ms)	Overlap Margin	Notes
Stem + Stage 1 + Stage 2	3.0	0	N/A	Triggers S3 prefetch
Stage 3	2.3	0.30	7.7× ✓	S3 streamed from DDR
Stage 4	1.8	0.69	3.3× ✓	S4 streamed from DDR
SPP	0.5	0	N/A	Permanent BRAM
PANet Neck	1.9	0	N/A	Permanent BRAM
Detection Head	0.9	0	N/A	Permanent BRAM
RoI + Projection	0.7	0	N/A	Permanent BRAM
Total compute	11.1	0.99 (overlapped)	—	217M MACs total
Control overhead	1.4	—	—	FSM + BRAM latency
VEGA task scoring	0.3	—	—	Software, 9,600 MACs
NMS-IP	0.2	—	—	Hardware pipeline
Total frame (theoretical)	13.0	—	—	~77 FPS
Total frame (conservative)	~18.0	—	—	~56 FPS

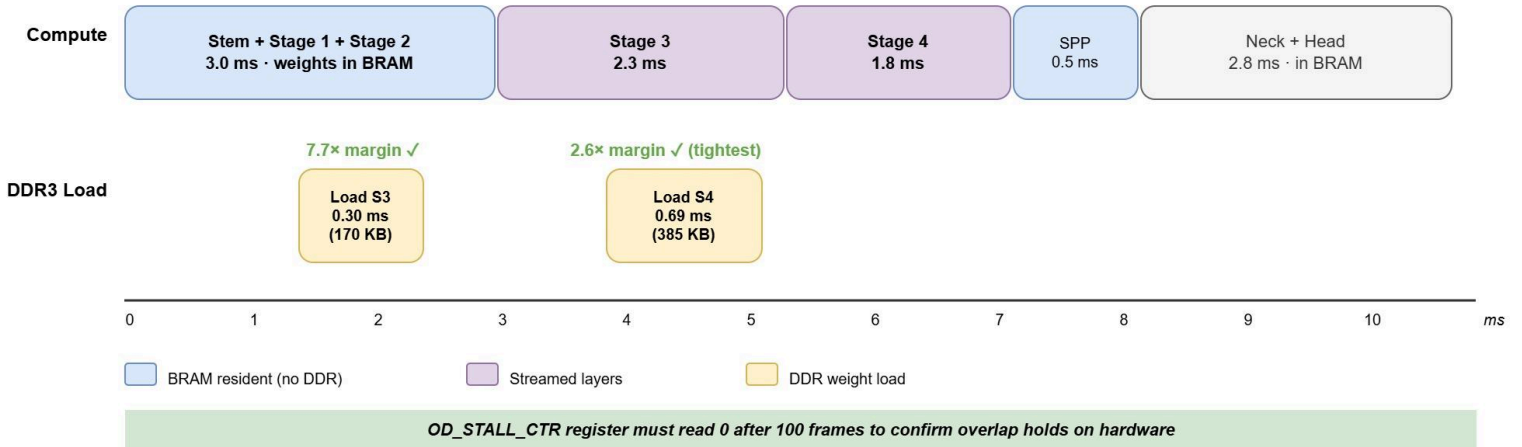
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Critical Path Analysis

At 150 MHz (6.67 ns period), four critical paths have been identified. The DSP48E1 cascade (PE-to-PE routing) is estimated at 3.5 ns (3.17 ns slack, high confidence). The BRAM-to-DSP path is estimated at 5.0 ns (1.67 ns slack, medium-high confidence), mitigated by the PBLOCK_WEIGHT_BRAM constraint keeping weight BRAMs adjacent to the systolic column. The AXI SmartConnect arbitration path is the tightest at an estimated 6.0 ns (0.67 ns slack, medium confidence), mitigated by the PBLOCK_VEGA constraint separating VEGA logic from DSP columns. A fallback to 125 MHz is defined if SmartConnect fails timing. The SAM-IP and NMS-IP paths are comfortable at 4.0 ns each (2.67 ns slack).

Figure 3: Weight Streaming Overlap Schedule

Scale: 1 ms = 80px | Origin x=160 | All DDR loads fully overlap compute — zero stall cycles



Rejected Design Alternatives

Rejected Design	Reason for Rejection
Custom TASKNet-FPGA backbone	168M MACs at 100% utilization, slower than YOLOv5n; requires 8+ weeks training from scratch
NanoDet-Plus (ShuffleNetV2)	Channel shuffle requires non-sequential BRAM addressing, +500 slices; estimated 28–32 ms FPGA latency vs 15 ms for YOLOv5n
MobileNetV3	Depthwise convolutions require separate 2,000-slice unit; early layer systolic utilization only 25–37%
TS-IP hardware decoders	4,500 slices + 25 BRAM36 to save 0.31 ms; negative ROI in 8-week timeline
10×10 systolic array	RTL complexity increase not justified; resolution reduction achieves same FPS gain more simply

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Rejected Design	Reason for Rejection
Hard-drop SAM gating	Creates unrecoverable false negatives; replaced by soft gating with configurable penalty factor
224×224 primary resolution	FPS budget does not close: 555M MACs / 19.2 GOPS = 28.8 ms sequential, cannot meet 50 FPS target

v. Testing and Implementation of Final Design

Simulation and Functional Verification

TACTILE's verification plan is structured around three testbench levels. Unit testbenches verify each IP block in isolation using cocotb (Python-based co-simulation). The NMS-IP unit testbench uses the canonical wine glass vs. beer glass test case: given beer glass (conf=0.81, task_score=0.18, class_prior=0.12) and wine glass (conf=0.72, task_score=0.94, class_prior=0.94) for task "serve wine", the fused scores are 0.017 and 0.636 respectively, and the wine glass must emerge as the top-ranked surviving proposal (Acceptance Criterion C4). The SAM-IP soft gate testbench injects a proposal whose center falls in a cold mask region and verifies that the effective confidence is attenuated by SAM_ALPHA (0.25) rather than dropped (Criterion C3).

Integration testbenches connect OD-IP, SAM-IP, and NMS-IP with a VEGA firmware simulation and verify the complete inference path on golden model outputs from the Python YOLOv5n reference implementation. A golden model comparison script validates that RTL outputs match software predictions to within quantization tolerance.

Hardware Implementation Plan

FPGA implementation follows a staged approach. Week 3 targets standalone VEGA AS1061 synthesis and timing closure, establishing the baseline resource budget. Week 4-5 targets OD-IP systolic array timing closure with mandatory Pblock constraints (PBLOCK_SYSTOLIC placing all 64 DSP48E1 in a single DSP column, PBLOCK_WEIGHT_BRAM keeping weight BRAMs adjacent to the systolic column, PBLOCK_VEGA separating processor logic from DSP regions). Week 6 targets end-to-end inference on hardware with FPS measurement via VEGA firmware timestamps from OD_CTRL START assertion to IRQ_DONE receipt. Week 7 removes ILA debug cores before final synthesis.

Acceptance Criteria

Gate	Criterion	Measurement Method
A1	$F_{max} \geq 150$ MHz on OD-IP post-place	Vivado timing report, WNS ≥ 0 on clk_pl
A2	$BRAM_{36} \leq 430$ without ILA	Vivado utilization report post-synthesis
A3	Slice utilization $\leq 80\%$ post-route	Vivado utilization report
B1	Measured FPS ≥ 50 at 160×160 on hardware	VEGA firmware timestamps, START to IRQ_DONE

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Gate	Criterion	Measurement Method
B2	OD_STALL_CTR = 0 across 100 frames	Read register after 100 consecutive frames
B3	VEGA task scoring ≤ 0.5 ms per frame	VEGA cycle counter around scoring loop
C1	SAM recall $\geq 97\%$ per task	Python evaluation on COCO-Tasks validation set
C2	End-to-end mAP $\leq 3\%$ worse than baseline	mAP@0.5 on COCO-Tasks test split
C3	Soft gate functional verification	Testbench: cold-region proposal has $\text{conf} \times 0.25$
C4	NMS-IP wine-glass test case correct	Unit testbench: wine glass beats beer glass

V. Results and Discussion

TACTILE's Stage 1 submission presents an architecture that has been fully specified at the RTL block level and validated through analytical performance modeling. Empirical hardware results will be reported in Stage 2 following FPGA implementation.

The analytical performance model predicts approximately 77 FPS theoretical and approximately 56 FPS conservatively at 160×160 resolution. The conservative figure accounts for DDR3 arbitration overhead, BRAM banking latency, and pipeline fill/flush cycles. This is well above the 50 FPS minimum requirement. The tightest margin in the design is the Stage 4 DDR streaming overlap ($3.3 \times$ compute-to-load ratio), which requires exclusive DDR3 access during inference — achieved by masking all VEGA interrupts except IRQ_SCORING and IRQ_DONE during the inference window.

The resource estimate of 33,880 slices (67.8%), 194 DSP48E1 (23.1%), and 421 BRAM36 (94.6%) without ILA meets all contest resource limits with comfortable headroom on slices and DSPs. BRAM is the tightest resource at $421/445 = 94.6\%$. A defined contingency exists: if post-synthesis BRAM count exceeds 430, the streaming buffer can be halved from 86 to 43 BRAM36, requiring two DDR load passes for Stage 4 and adding approximately 0.7 ms latency (FPS drops from ~ 56 to ~ 52), still above the 50 FPS gate.

The two novel contributions — SAM-IP spatial pre-filtering and NMS-IP fused scoring — address a fundamental limitation of task-agnostic detection pipelines. The SAM-IP's zero BRAM36 cost (all 199 weight bytes fit in LUTRAM) and parallel execution with image DMA mean that spatial task guidance is available before the first convolution at no latency cost. The NMS-IP fused score directly encodes the factored probability model of task-conditioned correct detection, providing a principled mathematical basis for the three-way multiplication rather than an ad-hoc weighting scheme.

Areas for future improvement include exploring higher input resolutions (192×192 or 224×224) if hardware FPS measurements in Stage 2 show FPS headroom beyond 50, and investigating whether the task embedding dimension can be reduced below 128 without significant mAP degradation (which would reduce VEGA scoring time and DDR3 proposal DMA overhead).

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The 646 spare DSP48E1 units provide the option to expand the systolic array to 10×10 if FPS falls short, but this is considered a last resort given the RTL complexity increase.

VI. Conclusion

TACTILE is a hardware-software co-designed edge AI system that addresses the task-driven object detection problem through three architectural novelties: a spatial task attention map that provides pre-inference region guidance at zero latency cost, an 8×8 INT8 systolic array with fully overlapped DDR3 weight streaming, and a three-way fused NMS that encodes task relevance, detector confidence, and class-task prior probability into a single ranking score.

The design targets the Genesys-2 FPGA board with the VEGA AS1061 RISC-V soft processor and meets all contest constraints: $\leq 50,000$ LUT slices (estimated 33,880), ≤ 430 BRAM36 (estimated 421 without ILA), $F_{max} \geq 150$ MHz, and ≥ 50 FPS (conservative estimate 56 FPS at 160×160). The hardware-software partitioning is justified analytically, with the VEGA software dot-product scoring loop and the PL accelerator pipeline forming a balanced co-design that exploits the strengths of both the RISC-V processor (flexible control, M-extension integer multiply) and the FPGA fabric (massive parallelism, deterministic memory access, DSP48E1 packing).

The significant contributions of this work are: (1) a quantified methodology for selecting and rejecting backbone architectures based on FPGA-specific efficiency metrics (systolic utilization, BRAM addressing overhead, DSP packing compatibility); (2) the SAM-IP architecture demonstrating that a meaningful spatial prior for task-driven detection can be computed using only 199 bytes of weights and 12 DSP48E1 slices; (3) the fused NMS formulation providing a theoretically grounded scoring rule for task-conditioned object selection; and (4) a complete resource and timing model enabling confident hardware implementation without design iteration.

VII. References

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